

Real-Time Driver Drowsiness Detection System Using Facial Landmark Analysis and Digital Signal Processing

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I. Abstract

This project proposes a real-time driver drowsiness detection system that uses facial landmark analysis and digital signal processing techniques to monitor driver alertness. The system captures live video from a laptop camera and processes facial landmarks using MediaPipe Face Mesh. Eye Aspect Ratio (EAR) is calculated from detected eye landmarks to measure eye closure behavior. A moving average filter is applied to smooth EAR values and reduce false detections caused by blinking or facial movement. When the EAR remains below a defined threshold for consecutive frames, the system identifies drowsiness and triggers audio and visual alerts. The project is designed for enclosed vehicles such as cars, buses, vans, and trucks. It runs offline using only a standard laptop, making it affordable, accessible, and practical for real-world implementation.

II. Background and Motivation

Driver fatigue is one of the major causes of road accidents worldwide. According to the World Health Organization, driver drowsiness contributes significantly to traffic-related injuries and fatalities. Long-distance drivers, night-shift workers, truck drivers, and public transport operators are particularly vulnerable.

Traditional drowsiness detection systems often require expensive sensors such as EEG monitors or specialized wearable devices. This project aims to provide a low-cost alternative using only computer vision and DSP techniques through a standard laptop camera.

The motivation of this project is to improve road safety by creating an affordable system capable of detecting early signs of driver fatigue in real time.

III. Objectives

- Develop a real-time facial monitoring system using MediaPipe Face Mesh
- Calculate Eye Aspect Ratio (EAR) for detecting eye closure
- Apply moving average filtering to reduce noise in EAR signals
- Implement threshold-based drowsiness detection logic
- Trigger audio and visual alerts when drowsiness is detected

IV. Technical approach with algorithm description

Step 1: Video Acquisition

Capture live video feed using OpenCV and laptop camera.

Step 2: Pre-processing

- Resize frame to 640x480
- Convert RGB format
- Perform ROI extraction
- Apply histogram equalization

Step 3: Facial Landmark Detection

Use MediaPipe Face Mesh to extract 468 facial landmarks.

Step 4: Eye Aspect Ratio Calculation

The system measures how open or closed the driver's eyes are by analyzing the distance between selected eye landmarks. A lower EAR value indicates that the eyes are closing, which may suggest drowsiness.

Step 5: Signal Processing

The collected EAR values are treated as a time-based signal. A moving average filter is applied to smooth sudden fluctuations caused by blinking, head movement, or camera noise. This helps reduce false detections.

Step 6: Decision Logic

If EAR remains below 0.25 for 20 consecutive frames:

→ Driver considered drowsy

Step 7: Alert System

- Audio warning

- On-screen visual alert

V. Tools and Resources Needed

Software	Hardware	Dataset
• Python • OpenCV • MediaPipe • NumPy • SciPy • Matplotlib • Pygame • GitHub	• Laptop (Core i5 minimum) • Built-in webcam • Speakers/headphones	• NTHU-DDD • UTA-RLDD

VI. Timeline with Milestones

Week	Dates	Milestone	Tasks	Hours/Week
1	Apr 28 – May 4	Environment Setup & Data Preparation	Install all Python libraries, configure OpenCV camera capture, and download the NTHU-DDD dataset. Verify that MediaPipe Face Mesh runs on the development machine at acceptable FPS.	8–10 hrs
2	May 5–11	Basic Signal Processing Pipeline	Implement EAR computation from the 6-point eye landmark subset. Log EAR values per frame to console and plot them over time using Matplotlib for analysis.	10–12 hrs
3	May 12–18	Core Algorithm Implementation	Add the 10-frame moving average filter to the EAR signal. Implement the 20-consecutive-frame threshold counter. Trigger a visual on-screen alert and audio alert through the laptop speakers when the condition is met.	12–14 hrs
4	May 19–25	Optimization & Testing	Evaluate system accuracy on the NTHU-DDD test set. Tune EAR threshold and frame-count parameters. Profile CPU usage and optimize the frame processing loop. Test in varied lighting conditions.	12–14 hrs

VII. Expected Deliverables

- Working drowsiness detection application
- Source code repository
- Performance evaluation report
- Final presentation slides
- Demonstration video

VIII. Anticipated Challenges and Solutions

Challenge	Anticipated Solution
Lighting variability reducing landmark detection accuracy	Apply histogram equalization in pre-processing; calibrate EAR threshold separately for bright and low-light conditions during testing
Normal blinks triggering false drowsiness alerts	Require the EAR threshold violation to persist for 20+ consecutive frames (~0.67 s); normal blinks last fewer than 10 frames (~0.33 s)
Eyeglasses or partially occluded eyes degrading EAR reliability	Implement a confidence check: skip frames where landmark confidence score falls below 0.5 and do not increment the drowsiness counter
Variation in EAR baseline across different drivers	Introduce a 5-second calibration phase at session start to compute a per-driver EAR baseline; set threshold as baseline minus an offset
Processing latency on lower-end laptops	Restrict MediaPipe input to 480p resolution; limit detection to face ROI bounding box; profile and remove bottlenecks using cProfile. Tested minimum spec: Intel Core i5, 4GB RAM.

IX. References

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